Machine Learning in Production
Midterm 1, Spring 2026

Claire Le Goues and Christian Kaestner

Name: ______________________________

Andrew ID: ______________________________

Instructions:

- Including this cover sheet and the scenario, your exam should have **9** pages. Make sure you are not missing any pages. *You may detach the last page and recycle it after the exam.*
- All questions in this midterm refer to the scenario on the last page. Answers are graded in the context of the scenario; **generic answers that do not relate to the scenario will not receive full credit.**
- The exam has a maximum score of **58** points. The point value of each problem is indicated. We designed the exam anticipating approximately one minute per point.
- **Please write legibly**. We are unlikely to be able to grade your solution if we can't read it.
- We give an amount of space commensurate with what we expect you to need for each question. We use horizontal lines to suggest where to not use the full page. You may exceed those limits if it is clear where to find the rest of your answer. However, we strongly recommend writing concise, careful answers; short and specific is much better than long, vague, or rambling.
- **Do NOT write anything you want us to grade on the back of pages.** We will scan the exam and will not look at the back sides.
- This is a **closed book exam**; no books or electronics allowed. You may refer to 6 sheets of notes (handwritten or typed, both sides). You do not need to hand in those sheets.

Question 1: Goals, Measurement and Telemetry [18p]

All questions in this exam relate to the QualityCheck project in the CardLens scenario on the last page. As a first step in the project, you want to make sure that your team is on the same page with what you want to achieve and how to measure whether you are successful.

(a) [8 points] State a *user goal* and a corresponding measure for the *QualityCheck* feature from the perspective of a card *buyer:*

User Goal:

Measure:

Data to collect (what and how):

Operationalization (computation):

(b) [6 points] You plan to evaluate the *HolographCheck* model of the *QualityCheck* feature *in production* with telemetry. You do not have direct access to the cards in the transactions, but you can make changes to the app and model to colect additional telemetry. You want to get a sense of how the model performs *before* cards are shipped. Describe a plausible approach to approximate model quality by observing the user's behavior that does NOT rely on prompting the users for explicit feedback.

Measure to approximate: *Average prediction accuracy of the HolographCheck model in the last 24h*

Data to collect (what and how):

Operationalization (computation):

(c) [4p] Describe two plausible *engineering challenges* when collecting telemetry data for evaluating the *HolographCheck* model in production (see Question 1b). No mitigation needed.

Engineering challenge 1:

Engineering challenge 2:

Question 2: Model, Data, and Infrastructure Testing [16p]

(a) [4p] You have 1000 videos of cards with corresponding metadata about the card and lighting stored in a database. You use this as training and test data for fine-tuning pretrained vision models for *QualityCheck*. You expect to regularly experiment and update the model, so you convert your notebook into a pipeline of several Python files.

Describe (with text or code sketch) one plausible useful test of this *training infrastructure* (not model or data) where it interfaces with another component of the system:

(b) [4p] The 1000 videos have been useful for experimentation and to estimate model accuracy, but you want to understand more specific behaviors about when the *CornerNet* model works well and when not. Select a specific behavior or quality you could test with the concept of **slicing** and describe how to test it without manually collecting additional data (text or code sketch). Your answer must demonstrate an understanding of the concept of slicing.

Specific model behavior/quality to evaluate:

How to test it using *slicing*:

(c) [4p] In addition to slicing, you consider property-based testing. Select a specific behavior or quality you could test with the concept of **property-based testing** and describe how to test it without manually collecting additional data (text or code sketch). Your answer must demonstrate an understanding of the concept of property-based testing.

Specific model behavior/quality to evaluate:

How to test it using *property based testing*:

(d) [4p] You are worried about how model quality may degrade over time. For the following concerns, distinguish whether they relate to concept drift (changes in decision boundaries), data drift (changes in input distributions) or schema drift (upstream changes in data encoding).

As cards become investment-grade assets, collectors apply increasingly strict corner wear thresholds. A corner that trained as quality 8/10 in 2021 now reflects 6/10 consensus.

☐ Concept drift ☐ Data drift ☐ Schema drift ☐ None of the above

CornerNet is retrained monthly on new labeled data. The labeling tool previously asked for a single corner wear score, but now asks labelers to provide separate ratings for whitening, fraying, and rounding, replacing the old corner wear score in the labels in the database.

☐ Concept drift ☐ Data drift ☐ Schema drift ☐ None of the above

Pokémon introduces a new generation of holofoil (hologram foil) with a different diffraction pattern and reflectance behavior under structured light. Your model now flags 30% of these authentic cards as potential counterfeits.

☐ Concept drift ☐ Data drift ☐ Schema drift ☐ None of the above

A new wave of high-quality counterfeit cards enters the market, using foil printing techniques that were previously unavailable to counterfeiters. HolographCheck's false negative rate on these cards is significantly higher.

☐ Concept drift ☐ Data drift ☐ Schema drift ☐ None of the above

Question 3: Risks and Mitigation [14 points]

To plan for mistakes you try to better understand the requirements and risks of the *QualityCheck* feature in the scenario. For this question, we consider the following *loss* (or harm or risk):

> *"An innocent seller unknowingly lists a counterfeit that the system validates with high condition grade and price, exposing the seller to fraud accusations."*

(a) [3p] State one **environmental assumption** that designers might make about the real-world context in which *QualityCheck* operates, but which is *likely not actually true* in practice, and could therefore contribute to the loss above.

(b) [3p] Your team proposes several system-level mitigations for this issue. For each, indicate whether it *reduces* the likelihood of loss (improved reliability), or *prevents* the loss (guarantee), or *neither.* (select an option, no justification needed)

Mitigation 1: Add a disclaimer on every listing stating that QualityCheck does not guarantee card authenticity.

☐ reduces the likelihood of loss ☐ prevents the loss ☐ neither

Mitigation 2: Flag listings where the card's print year is more than 15 years ago and the condition grade is 9/10 or above for review by a QualityCheck (human) staff member and domain expert.

☐ reduces the likelihood of loss ☐ prevents the loss ☐ neither

Mitigation 3: Use a multimodal LLM as a second-opinion check on listings above $50, prompting it with the card photo and asking whether the card appears authentic.

☐ reduces the likelihood of loss ☐ prevents the loss ☐ neither

(c) [4p] Considering cost and effectiveness and the principle of *"as low as reasonably practicable"* in risk management, which of the proposed mitigation would you recommend the company prioritizes most and least? Briefly justify each answer.

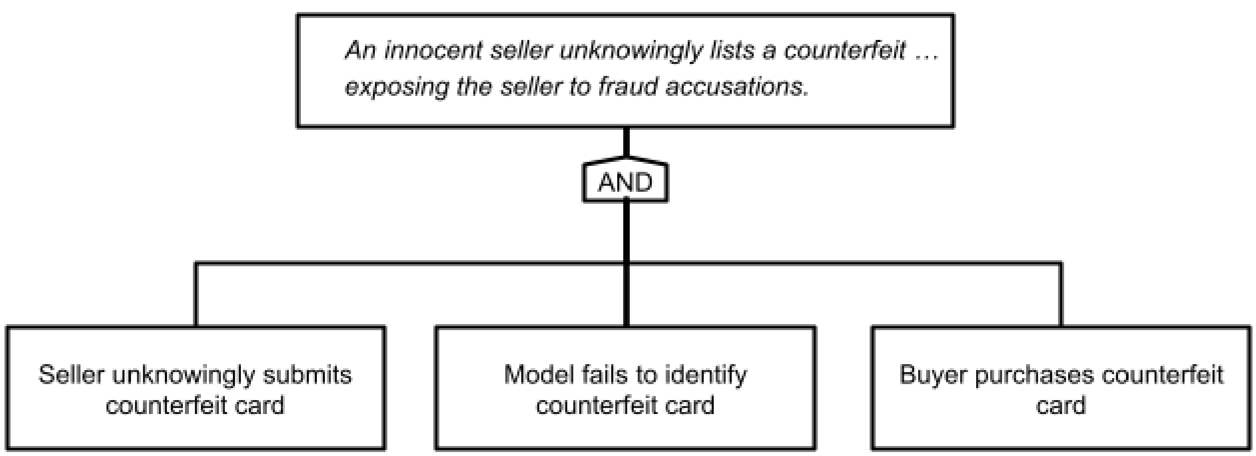
Highest priority: ☐ 1 (disclaimer) ☐ 2 (selective review) ☐ 3 (LLM)

Justification:

Lowest priority: ☐ 1 (disclaimer) ☐ 2 (selective review) ☐ 3 (LLM)

Justification:

(d) [4 points] Update the fault tree below to reflect *mitigation 2 (selective review)* from above. You can cross out nodes/ connections or add additional nodes/connections as necessary.



Question 4: Deployment and Scaling [10 points]

(a) [6p] All past models of the CardLens system (for analyzing photos and predicting prices) were hosted in the cloud. You are considering whether to deploy the *QualityCheck* models in the cloud or in the app on the user's device. List an argument for each and make and justify a recommendation. (keywords are sufficient)

Argument for cloud deployment:

Argument for on-device deployment:

Recommended deployment strategy: ☐ Cloud ☐ On device (in the mobile app)

Justification:

(b) [4p] Independently from your recommendation above, you are considering how to scale the system if you were to deploy the model in the cloud. Which of the following statements are technically correct (no justification needed)?

Stream processing would be a good choice, because each video is processed immediately as a continuous data flow with sub-second latency.

☐ Correct ☐ Incorrect

If we want to occasionally reanalyze all videos for testing new models, batch processing is a good choice, because the models are probably smaller than the videos.

☐ Correct ☐ Incorrect

The lambda architecture allows us to avoid the complexity of batch processing, to instead perform inference just as needed.

☐ Correct ☐ Incorrect

Deploying the models as a microservice is a bad idea because it would force us to analyze videos sequentially (only one video at a time) and thus create a bottleneck.

☐ Correct ☐ Incorrect

Scenario: Quality Checks for a Trustworthy Trading Card Marketplace

CardLens is a small but very successful company (12 employees on the tech side) operating a mobile app and web marketplace for card game collectors, focusing on Pokémon, Magic: The Gathering, and Yu-Gi-Oh!. New cards are released regularly, but there is a multi-billion dollar secondary market. Card condition is the primary driver of price, alongside rarity; professional grading services exist but are slow and expensive. CardLens generates revenue through transaction fees, and backs every sale with a money-back guarantee on condition accuracy, making grading precision a direct financial concern.

Fig. 1: a satisfied customer.

The company's existing product is effective at both card identification and price prediction based on historical sales. However, sellers self-report card condition, which has proven increasingly unreliable as the marketplace scales. Dispute rates are climbing and the money-back guarantee creates increasing liability exposure.

The company is now developing **QualityCheck**, a new system to replace seller-reported condition with AI-produced grades. Rather than analyzing a single photograph, QualityCheck uses a structured light capture protocol: The user places a card on a table in a dark room, and the app cycles the phone's screen through a sequence of lighting conditions while the front camera records video and the user moves the phone around. This produces approximately 40MB of synchronized video with known illumination properties and angles.

You are currently working on two quality checks, using different models:

- **CornerNet**, a model assessing *corner wear*: A card with perfect surfaces but damaged corners (e.g., whitening or rounding) cannot achieve a top grade. Corner defects create characteristic shadows under raking light (light shone at a sharp angle).
- **HolographCheck**: a model assessing the condition of holographic foil layers on premium cards, which can indicate damage or counterfeits. The model analyzes paper reflectance, print dot patterns, and color spectrum under the structured lighting protocol.

The quality check models have infrastructure to process the data. For example, CornerNet preprocessing extracts 128×128 pixel crops of each corner from each lighting condition, ultimately outputting a wear score (0–10) per corner, a confidence estimate, and a wear-type classification. HolographCheck, in turn, returns an authenticity score.

You have over half a million photos of cards from past operations and scraped from the internet (e.g., eBay), not always with reliable metadata. In addition, an intern has taken about 1000 videos of cards with known conditions (i.e., fairly reliable labels). The proprietary capture format means no external datasets exist, and all training data must be collected through the app itself. Most of your video data is on English cards.

CardLens currently serves 60,000 daily active users and you expect over 500,000 quality checks a day eventually (about 6 per second).